

FEASIBILITY STUDY FOR AQUASENSE

1. INTRODUCTION

The feasibility study for AquaSense evaluates the practicality and viability of developing a holistic Smart Water Supply and Management Platform. This study covers both the Mini Project Phase (Digitization of Billing, Complaints, and Connections) and the Main Project Phase (IoT-based Leak Detection, AI Automation, and Personal Hydration Tracking).

This assessment analyzes Economic, Technical, and Behavioral factors to ensure the project can be successfully deployed using the proposed technology stack (Flutter, Firebase, ESP32 IoT, Python AI) within the academic timeline.

2. ECONOMIC FEASIBILITY

Assessment

The project is economically viable due to the reliance on open-source technologies and scalable cloud infrastructure that minimizes upfront capital expenditure.

Software Costs: Development utilizes Flutter and Dart, which are free and open-source. The backend runs on Google Firebase. The "Blaze Plan" (Pay-as-you-go) is required for Cloud Functions (needed for the Main Project's AI/ML services), but the free tier limits are sufficient for development and testing, keeping operational costs near zero during the academic phase.

Hardware Costs:

Required only basic ESP32 setup (~₹1,000).

Adds water flow sensors (YF-S201) and pressure transducers. The estimated total cost for a functional prototype node is approximately ₹3,500 - ₹4,500. This is significantly cheaper than industrial SCADA systems.

Return on Investment (ROI): The Main Project's Leak Detection System directly addresses "Non-Revenue Water" (water lost before billing). By detecting leaks in real-time, the system saves the utility provider significant money in wasted resources and pumping costs.

Value Addition: The new Hydration Module adds lifestyle value to the application without any additional hardware cost, increasing the app's "stickiness" and user retention.

Conclusion: The project is economically feasible as the low development costs are justified by the potential operational savings and service improvements.

3. TECHNICAL FEASIBILITY

Assessment

The technical requirements for both phases are achievable given the maturity of the chosen modern tech stack.

Mobile Application (Frontend): Flutter provides a robust environment for building the complex UI required for the Hydration Tracker (animations) and Dashboard (real-time charts). Libraries for Local Notifications (`flutter_local_notifications`) and background processing are mature and stable.

IoT Integration (Main Project):

The integration of ESP32 with Firebase Realtime Database/Firestore is a well-documented pattern.

Simulating leak detection using logic comparison (Source Flow vs. Destination Flow) is technically straightforward to code in C++ for the microcontroller.

AI & ML (Main Project):

Complaint Categorization: Using pre-trained models (like MobileNet) or APIs (Google Cloud Vision) via Cloud Functions is a standard industry practice.

Predictive Analytics: Python scripts running on Cloud Functions can easily handle the regression algorithms needed for demand forecasting.

Hydration Logic: The logic for calculating daily intake (based on weight/gender) and scheduling notifications is computationally simple and runs locally on the device, posing no technical risk.

Conclusion: The development team possesses the necessary skills in Flutter and Firebase. The addition of IoT and ML introduces complexity, but it is manageable within the scope of a Main Project.

4. BEHAVIORAL FEASIBILITY

Assessment

This factor analyzes user acceptance. The project is designed to solve specific pain points for all stakeholders.

For Citizens:

Utility: Online billing and complaint tracking remove the frustration of queuing and unanswered phone calls.

Lifestyle Integration (New): The Hydration Module transforms the app from a monthly utility tool into a daily lifestyle companion. This addresses the "forgetfulness" factor in drinking water, providing immediate personal health benefits.

For Supervisors:

Workload Reduction: Automated leak alerts (Main Project) replace the need for constant manual patrolling. QR-based billing speeds up the monthly reading cycle.

Trust: AI-categorized complaints prevent ambiguity regarding the severity of reported issues.

Adoption Risks: There may be initial resistance from older supervisors to trust "AI predictions" or "Sensor Alerts" over physical inspection. This is mitigated by keeping the system advisory—supervisors still make the final approval.

Conclusion: The system is behaviorally feasible as it enhances convenience and health for citizens while automating tedious tasks for staff.

5. FEASIBILITY STUDY QUESTIONNAIRE

Q1. What is the core purpose of AquaSense?

A: To digitize water management, ensuring transparency in billing, real-time leak detection, and promoting water conservation.

Q2. How are leaks currently detected (Existing System)?

A: Reactive detection. We rely on citizens seeing water on the road and reporting it via WhatsApp or phone calls, often hours or days after the leak starts.

Q3. How will the Main Project improve leak detection?

A: By integrating Flow and Pressure sensors. If the pressure drops abnormally or outflow exceeds inflow, the system triggers an instant alert to the Supervisor, enabling a proactive response before water is wasted.

Q4. How does the current billing system work?

A: Manual entry in logbooks and cash collection. It is slow, prone to error, and lacks a digital audit trail.

Q5. How will AquaSense handle billing?

A: Supervisors scan a QR code on the meter to fetch user details instantly. The bill is calculated automatically, and citizens can pay online via Razorpay.

Q6. What is the benefit of the new "Hydration Module" for a utility app?

A: It increases citizen engagement. While billing happens only once a month, water drinking is a daily activity. This feature encourages users to keep the app installed and active, promoting the brand message of "Health & Conservation."

Q7. How does the AI Complaint Categorization help Supervisors?

A: The AI analyzes the user's photo and auto-tags the severity (e.g., "Critical - Pipe Burst"), allowing supervisors to prioritize urgent repairs immediately.

Q8. Is the cost of the new IoT hardware justified?

A: Yes. A single major pipe burst can waste lakhs of liters of treated water. The cost of a sensor node (approx. ₹4,000) is recovered quickly by preventing just a few such incidents.

Q9. Are citizens ready to use a mobile app for these services?

A: Yes, smartphone penetration is high. The convenience of paying bills from home and getting transparency on complaints is a strong motivator for adoption.

Q10. What if the internet connectivity fails for the IoT sensors?

A: The ESP32 nodes can be programmed to cache data locally and upload it once connectivity is restored.